

Tactic Zone

Application Documentation

Fields, Features, Roles & Data Structures

This document describes all modules of the Tactic Zone platform:

1. Tactic Zone Analytics (coaching dashboard)
2. FIFA World Cup 2030 portal (bookings, Pi payments, Fan Passport)
3. Backend API and CSV data pipeline

Includes form fields, user roles, feature matrices, and demo credentials.

1. Global Features Overview

The platform combines football analytics (computer vision + AI coaching) with a FIFA World Cup 2030 fan experience (hotels, flights, tickets, Pi Network payments, and NFT-inspired Fan Passport collectibles).

Feature	Module	Description
Marketing site	Main	Landing, About, pricing, testimonials
Client-side auth	Main + WC	Demo users + localStorage; no server auth
Match analytics	Main	15+ pages backed by CV CSV pipeline
Video upload pipeline	Main	Upload files/URLs -> CV + recommendations + reports
AI Coach Assistant	Main	LLM chat grounded in match context
Coach Reports	Main	5 report types; full/halftime modes
Tactical Puzzles + XP	Main	Gamification with levels and accuracy
Live pipeline monitor	Main	Real-time CV progress
World Cup 2030 portal	WC	Hotels, flights, tickets, packages, teams
Pi Network payments	WC	USD or Pi checkout (demo wallet)
Fan Passport	WC	Auto-mint collectibles from bookings
i18n + theme	WC	EN/FR/AR/ES/PT; dark/light mode

2. Tactic Zone Main App

2.1 User Roles

Role	Access	Notes
admin	Full dashboard	Demo account; identity label only
coach	Full dashboard	Default for new registrations
analyst	Full dashboard	Demo account; same access as admin/coach
Guest	Marketing + auth pages	No /app or /coach access

Note: All three logged-in roles see identical navigation. ProtectedRoute only checks login status.

2.2 User Model Fields

Field	Type	Description
email	string	Login identifier
name	string	Display name
club	string	Club/team affiliation
role	enum	admin coach analyst
country	string	Optional user country
coachingLevel	string	Grassroots to International

2.3 Routes

Route	Page	Auth
/	HomePage	Public
/about	AboutPage	Public
/login	LoginPage	Public
/register	RegisterPage	Public
/forgot-password	ForgotPasswordPage	Public
/reset-password	ResetPasswordPage	Public
/coach	CoachHubPage	Protected
/app	OverviewPage	Protected
/app/match	MatchSummaryPage	Protected
/app/players	PlayerStatsPage	Protected
/app/squads	SquadBreakdownPage	Protected
/app/physical	PhysicalPerformancePage	Protected
/app/events	EventsFormationsPage	Protected
/app/tracking	TrackingExplorerPage	Protected
/app/recommendations	RecommendationsPage	Protected
/app/reports	ReportsPage	Protected
/app/assistant	AssistantPage	Protected

/app/puzzles	PuzzlesPage	Protected
/app/upload	UploadPage	Protected
/app/news	NewsPage	Protected
/app/live	LivePage	Protected
/app/tactical	TacticalPage	Protected
/app/timeline	TimelinePage	Protected
/world-cup/*	WorldCupApp	Separate WC auth

3. Tactic Zone - Form Fields

3.1 Authentication Forms

Form	Field	Type	Validation
Login	Email	email	Required
Login	Password	password	Required
Register	Full Name	text	Required
Register	Club / Team Name	text	Required
Register	Email	email	Required, unique
Register	Password	password	Min 8 characters
Register	Confirm Password	password	Must match
Register	Country	select	Morocco, France, Spain, UK, USA, Brazil, Other
Register	Coaching Level	select	Grassroots, Academy, Semi-Pro, Pro, International
Forgot Password	Email	email	Must exist in user store
Reset Password	New Password	password	Min 8 characters
Reset Password	Confirm Password	password	Must match

3.2 Functional Forms & Controls

Page	Field	Type	API / Notes
Upload	Team video file	file	team_video multipart
Upload	Opponent video file	file	opponent_video multipart
Upload	Team video URL	url	team_url
Upload	Opponent video URL	url	opponent_url
Upload	Pipeline mode	select	full halftime live_lite
Reports	Report mode	select	full halftime
Assistant	Context	buttons	match, opponent, both, training, tactical
Assistant	Message	text	POST /api/assistant/chat
Puzzles	Category	select	From API categories
Puzzles	Difficulty	select	Easy, Medium, Hard, Elite
Puzzles	Answer	A/B/C/D	POST puzzle/submit
Player Stats	Team filter	select	team: 1 or 2
Player Stats	Min distance	number	min_distance filter
Tracking	Frame	range	frame index
Tracking	Class	select	player ball
Header	Match selector	buttons	video_id: 1, 2, or both

4. World Cup 2030 Module

4.1 User Roles

Role	Dashboard	Special Access
fan	FanDashboard	Passport, favorites, donations, preferred team
hotel_manager	HotelManagerDashboard	Listings, rooms, pricing (UI)
travel_agency	AgencyDashboard	Packages, tours, offers (UI)
admin	AdminDashboard	Passport admin, QR validation, analytics

4.2 WC User Model Fields

Field	Type	Description
email	string	Login identifier
name	string	Display name
role	enum	fan hotel_manager travel_agency admin
country	string	Default Morocco on register
favorites	string[]	Team IDs (fans only) e.g. mar, esp
preferredTeam	string	Team ID for recommendations

4.3 WC Routes

Route	Page	Auth
/world-cup	Home	Public
/world-cup/hotels	Hotels	Public
/world-cup/flights	Flights	Public
/world-cup/tickets	Tickets	Public
/world-cup/packages	Packages	Public
/world-cup/teams	Teams & Stats	Public
/world-cup/stadiums	Stadiums	Public
/world-cup/login	Login	Public
/world-cup/register	Register	Public
/world-cup/dashboard	Dashboard	WC login required
/world-cup/passport	Fan Passport	Fan role only

5. Bookings, Payments & Passport

5.1 Booking Record Fields

Field	Type	Description
id	string	Auto: BK-{timestamp}
type	enum	hotel flight ticket package donation
title	string	Display title
date	string	YYYY-MM-DD
price	number	USD amount
paymentMethod	enum	usd pi
piAmount	number	Pi amount if crypto payment
status	enum	confirmed pending
details	string	Rich description (stadium, amenities, etc.)

5.2 Donation Record Fields

Field	Type	Description
id	string	Auto: DN-{timestamp}
target	string	Team or coach name
targetType	enum	team coach
amount	number	USD (min 5)
paymentMethod	enum	usd pi
piAmount	number	Pi equivalent
date	string	ISO date

5.3 Booking Page Search Fields

Page	Fields
Hotels	City (select), Max price (slider \$100-1000), Min stars (3+/4+/5+)
Flights	From, To (airport codes), Depart date, Passengers (1/2/3+)
Tickets	Category 1 / 2 / 3 per match
Packages	Package selection with price
Fan Donations	Target type (team/coach), Recipient select, Amount USD

5.4 Pi Network Payment (Demo)

- Exchange rate: ~45 Pi per 1 USD
- Connect button simulates wallet; username: pioneer_fan
- PaymentModal: toggle USD vs Pi; auto-connect on Pi checkout
- Storage keys: wc2030_pi_connected, wc2030_pi_user

5.5 Fan Passport Fields

Field	Type	Description
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passportId	string	WC2030-PAS-XXXX-XXXX unique per user
userEmail	string	Owner email
loyaltyPoints	number	From rarity points + badges
collectibles	array	Digital wallet items
badges	array	12 achievement badges
visitedCities	array	From bookings
visitedStadiums	array	From ticket collectibles
matchCount	number	Ticket bookings count

5.6 Collectible Card Fields

Field	Type	Description
id	string	COL-{timestamp}-{segment}
bookingId	string	Source booking
type	enum	ticket hotel flight package
rarity	enum	common rare epic legendary
serialNumber	string	WC2030-XXXX-XXXX unique
title / subtitle	string	Match or booking info
stadium	string	Match tickets only
city / country	string	Host location
date	string	Booking date
seatNumber	string	Generated for tickets e.g. A12-34
teamHome / teamAway	string	Team flags from match title
qrPayload	JSON	QR code for stadium validation
validated	boolean	Entry validated at gate

5.7 Rarity & Loyalty Points (Default)

Rarity	Ticket Triggers	Points
Legendary	Final, Semi-Final, Category 1, VIP packages	1000
Epic	Category 2, travel packages	400
Rare	Default tickets; 5-star hotels	150
Common	Flights, standard hotels	50

6. Per-Role Feature Matrix

6.1 Tactic Zone Main App

Feature	Admin	Coach	Analyst	Guest
Marketing pages	Yes	Yes	Yes	Yes
Coach Hub /app/*	Yes	Yes	Yes	No
Upload pipeline	Yes	Yes	Yes	No
AI Assistant + Reports	Yes	Yes	Yes	No
Puzzles + XP	Yes	Yes	Yes	No

6.2 World Cup 2030

Feature	Fan	Hotel Mgr	Agency	Admin	Guest
Browse catalog	Yes	Yes	Yes	Yes	Yes
Book + pay USD/Pi	Yes	Yes	Yes	Yes	Yes*
Fan Dashboard	Yes	No	No	No	No
Fan Passport page	Yes	No	No	No	No
Donations	Yes	No	No	No	No
Collectible auto-mint	Yes	Yes	Yes	Yes	No
Passport Admin + QR	No	No	No	Yes	No
Notifications bell	Yes	Yes	Yes	Yes	No

*Bookings work without login (localStorage). Dashboard/passport require login.

7. Backend API Endpoints

7.1 Analytics (video_id: 1 or 2)

Method	Endpoint	Purpose
GET	/api/status	Pipeline + analytics readiness
GET	/api/analytics/overview	Cross-match summary
GET	/api/analytics/matches/{id}	Team match statistics
GET	/api/analytics/players/{id}	Player-level CV metrics
GET	/api/analytics/squads/{id}/{team}	Squad breakdown
GET	/api/analytics/tracking/{id}	Frame-by-frame tracks
GET	/api/analytics/tactical/{id}	Thirds, width, compactness
GET	/api/analytics/timeline/{id}	Match event timeline

7.2 Recommendations, Reports, Assistant

Method	Endpoint	Purpose
GET	/api/recommendations/formations	Formation recommendations
GET	/api/recommendations/lineup	Recommended lineup
GET	/api/recommendations/squads	Squad player data
GET	/api/reports	Cached coach reports
POST	/api/reports/generate	Generate LLM reports
POST	/api/assistant/chat	AI coach chat
GET	/api/gamification/profile	XP, level, accuracy
POST	/api/gamification/puzzle/submit	Submit puzzle answer
POST	/api/pipeline/upload	Upload match videos
POST	/api/pipeline/start	Start CV pipeline

7.3 Key CSV Output Columns

File	Key Columns
teams_final_statistics	formations, score, goals, possession, passes, tackles
player_statistics	shirt_number, Position, Goals, Distance_covered, Avg_speed
tracks_csv	Frame Number, Class Label, Track ID, Bounding Box
recommended_ formations	formations, score, defenders, midfielders, attackers
combined_team	position, Shirt_Number, status (Starting 11 / Substitute)
closest_player_data_mobile1	Morocco squad stats per player
morocco_roster	Player_Name, Position, Image_File, Key_Star

8. Demo Credentials

8.1 Tactic Zone Main App

Role	Email	Password
Admin	admin@tacticzone.com	Admin123!
Coach	coach@tacticzone.com	Coach123!
Analyst	analyst@tacticzone.com	Analyst123!

8.2 World Cup 2030

Role	Email	Password
Fan	fan@wc2030.com	Fan123!
Hotel Manager	hotel@wc2030.com	Hotel123!
Travel Agency	agency@wc2030.com	Agency123!
Admin	admin@wc2030.com	Admin123!

8.3 Quick Start

- Backend: `python -m uvicorn api_server:app --host 127.0.0.1 --port 8000`
- Frontend: `cd dashboard && npm run dev`
- Main app: `http://localhost:5173`
- World Cup: `http://localhost:5173/world-cup`
- API docs: `http://localhost:8000/docs`

8.4 Local Storage Keys

Key	Module	Contents
tacticzone_auth_user	Main	Current session user
wc2030_user	WC	Current WC session
wc2030_bookings	WC	All bookings
wc2030_passports	WC	Fan passports + collectibles
wc2030_pi_connected	WC	Pi wallet connection state